

chap_simples PLAN SI

2025-07-07

Détermination de la taille d'échantillon

```
e <- 0.03
p <- c(0.05, 0.10, 0.3, 0.5)
z <- qnorm(0.975,0,1)
(n0 <- z^2*p*(1-p)/e^2)
```

```
## [1] 202.7437 384.1459 896.3404 1067.0719
```

```
#[1] 202.7511 384.1600 896.3733 1067.1111
```

```
e <- 0.01
(n0 <- z^2*p*(1-p)/e^2)
```

```
## [1] 1824.693 3457.313 8067.064 9603.647
```

```
#[1] 1824.76 3457.44 8067.36 9604.00
```

Plan SI

```
set.seed(123456)
sample(20,5)
```

```
## [1] 10 20 17 7 13
```

```
#[1] 16 15 8 6 17
```

```
library(sampling)
set.seed(123456)
(si <- srswor(5,20))
```

```
## [1] 0 0 0 0 0 0 1 0 0 1 0 0 1 0 0 0 1 0 0 1
```

```
#[1] 0 0 0 0 0 1 0 1 0 0 0 0 0 0 1 1 1 0 0 0
```

```
which(si==1)
```

```
## [1] 7 10 13 17 20
```

```
#[1] 6 8 15 16 17
```

```
set.seed(345678)
srswor(5,20)
```

```
## [1] 0 1 0 1 0 0 0 0 0 1 0 0 0 0 0 0 1 0 1 0
```

```
#[1] 0 0 0 1 0 0 1 0 0 0 0 0 1 0 1 0 1 0 0 0
```

```
set.seed(123456)
srswor(5,20)
```

```
## [1] 0 0 0 0 0 0 1 0 0 1 0 0 1 0 0 0 1 0 0 1
# [1] 0 0 0 0 0 1 0 1 0 0 0 0 0 0 1 1 1 0 0 0

w <- writesample(5,20)
dim(w)

## [1] 15504      20
# [1] 15504
# 20

library(sampling)
data(rec99)
(N <- nrow(rec99))

## [1] 554
# [1] 554

n <- 70
set.seed(123456)
si.rec <- srswor(n,N)
(nom <- rec99$COMMUNE[si.rec==1][1:10])

## [1] "AIGNES"      "AIGREFEUILLE" "ARBAS"      "ARBON"      "ARDIEGE"
## [6] "AURIBAIL"     "BAREN"        "BAZIEGE"    "BENQUE-DESSO" "BOULOGNE-SUR"
# [1] CAZAUNOUS MILHAS ASPET AURAGNE BEAUMONT-SUR
# [6] BENQUE-DESSO JURVIELLE SALLES-ET-PR TREBONS-DE-L AIGREFEUILLE

si.logvac <- rec99$LOGVAC[si.rec==1]
N*mean(si.logvac)

## [1] 14388.17
# [1] 9441.743

pik.si <- rep(n/N, n)

HTestimator(si.logvac,pik.si)

##           [,1]
## [1,] 14388.17
# [,1]
# [1,] 9441.743

set.seed(345678)
si.rec2 <- srswor(n,N)
si.logvac2 <- rec99$LOGVAC[si.rec2==1]
HTestimator(si.logvac2,pik.si)

##           [,1]
## [1,] 8507.857
# [,1]
# [1,] 10383.54

HTestimator(si.logvac,pik.si)-sum(rec99$LOGVAC)

##           [,1]
```

```
## [1,] 3620.171
#[,1]
#[1,]-1326.257
HTestimator(si.logvac2,pik.si)-sum(rec99$LOGVAC)

##          [,1]
## [1,] -2260.143
# [,1]
# [1,]-384.4571
N^2*(1-n/N)*var(si.logvac)/n

## [1] 5800506
#[1] 1745875
pikl.si <- matrix(n*(n-1)/(N*(N-1)),n,n)
diag(pikl.si) <- n/N
varHT(si.logvac,pikl.si)

## [1] 5800506
# [1] 1745875
library(survey)

## Loading required package: grid
## Loading required package: Matrix
## Loading required package: survival
##
## Attaching package: 'survival'
## The following objects are masked from 'package:sampling':
##
##   cluster, strata
##
## Attaching package: 'survey'
## The following object is masked from 'package:graphics':
##
##   dotchart
ech.si <- svydesign(id=~CODE_N, weights=rep(N/n,n),
  fpc=rep(n/N,n), data=rec99[si.rec==1,])
(est.si <- svytotal(~LOGVAC, ech.si)[1])

## LOGVAC
## 14388.17
# LOGVAC
# 9441.743
(se.si <- SE(svytotal(~LOGVAC, ech.si)))

## LOGVAC
## LOGVAC 2408.424
```

```
# LOGVAC
# LOGVAC 1321.316
```

```
svytotal(~LOGVAC, ech.si)
```

```
##          total      SE
## LOGVAC 14388 2408.4
```

```
#total
#SE
#LOGVAC 9441.7 1321.3
```

```
attributes(ech.si)
```

```
## $names
## [1] "cluster"      "strata"      "has.strata" "prob"      "allprob"
## [6] "call"         "variables"   "fpc"        "pps"
##
## $class
## [1] "survey.design2" "survey.design"
```

```
#$names
#[1] "cluster"
#"strata"
#[6] "call"
#"has.strata" "prob"
#"variables" "fpc"
#"allprob"
#"pps"
```

```
ech.si$variables[1:5,]
```

```
##      CODE_N      COMMUNE BVQ_N POPSDC99 LOG LOGVAC STRATLOG surf_m2 lat_centre
## 2      31002      AIGNES 31033      193 99      7      1 22017258 43.33456
## 3      31003 AIGREFEUILLE 31044      577 179      1      2 4608949 43.56919
## 11     31011      ARBAS 31523      186 230      7      2 7327314 42.99228
## 12     31012      ARBON 31483       76 79      7      1 4448211 43.00065
## 13     31013      ARDIEGE 31483      325 156      8      2 3802439 43.06678
##      lon_centre
## 2      1.5866078
## 3      1.5851581
## 11     0.9054543
## 12     0.7317850
## 13     0.6445605
```

```
#CODE_N
#COMMUNE BVQ_N POPSDC99 LOG LOGVAC stratlog
#2 8 31131 31342 10 31020 42 31024 CAZAUNOUS 31020 MILHAS 31020 ASPET 31020 47 56
#4 140 150 923 684 AURAGNE 31033 43 31052 BEAUMONT-SUR 31033 310 129
#1414 563 10 90 8 20 1 2 3 2 3
```